How does a SVM choose its support vectors?

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Lets say, we are given a training dataset of

n points of the form

$$(\vec{x}_1, y_1), \ldots, (\vec{x}_n, y_n)$$

A support vector is a vector of a datapoint xi that lies on the hyperplane(s)

In order to choose the support vectors, we want to maximize the margin m and that implies we reduce the magnitude or norm of the vector that's perpendicular to the hyperplanes(s) and closest to a datapoint.

We know from the margin calculation formula that

m=2 / ||w||

which implies that the lower the norm of vector w, then greater is the margin.

But to identify the margin we need to make sure all the vectors of all datapoints to satisfy the SVM constraint.

yi... (more)

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In layman's terms, support vector machine is a generalization of Nearest Neighbor (NN) algorithm. So let us understand NN first (If you already know about NN, you can directly jump to SVM part below).

Nearest Neighbor Algorithm

It is a very simple algorithm. You are given a training data consisting of m training instances $\{(x^1, y^1), (x^2, y^2), \ldots, (x^m, y^m)\}$.

Here y^i is the class label of i^{th} instance. For example, in the figure below, each instance can have either label 1 (blue points), or label -1 (red points).

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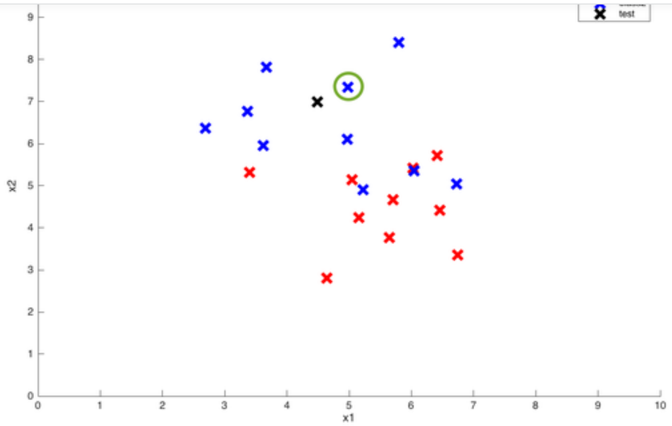


For now, let us assume we have two possible classes 1 or -1.

Now you are given a test point (black point above), and you have to predict its class, whether it belongs to red class or blue class. In NN algorithm, you find the nearest training point to this test point (by measuring distance of test point from **every** training point), and the class predicted of test point is same as of nearest training point. From now on, we will use the word **similarity** instead of distance. Lower the distance, higher the similarity between two points. This will be useful to understand SVMs. For example, the circled point is closest to test point, and hence class of test point is blue.

So some of the observations from NN algorithm are :

1. There is no training phase in this algorithm. We don't do any computation alone with training points. Only when a test point comes, we compute similarity from **every** training point. This is a big disadvantage of NN algorithm. Consider having millions of training points, and for every test point, we have to calculate millions of similarities from test point.
2. We calculate similarities from the training points which are very very far from test points, which are not really required.
3. We don't give any importance to other training points except the nearest one.

Support Vector Machine

In SVM, we remove these disadvantages as :

1. Instead of finding similarity from every training point, we calculate similarity from only a **subset** of training points, which we compute in the **training phase**. So in the training phase, we fix a subset of training points from which we compute the similarity of any test point. These selected training points are called **support vectors,** since only these points will support our decision of selecting the class of a test point. Our hope is that our training phase finds as few as support vectors so that we have to compute fewer number of similarities.
2. Now once we have selected support vectors, we assign a **weight** to each support vector, which basically tells how much importance we want to give to that support vector while making our decision. So unlike NN, we don't give importance to only a single training point, instead we give each support vector a separate importance.

Now how do we take the decision in this setting? We just compute a weighted sum of similarities from test point to each of the support vector and compute the class based on this weighted sum.

More concretely, suppose out of m training instances, we have selected k instances as support vectors. Let these support vectors are $\{(s^1, y^1), (s^2, y^2), \ldots, (s^k, y^k)\}$, and let their weights are $\{\alpha^1, \alpha^2, \ldots, \alpha^k\}$. Let x is a test point. Now we compute weighted sum of similarities from x to each of the support vectors :

$D = \sum_{j=1}^{k} y^j * \alpha^j * sim(x, s^j)$

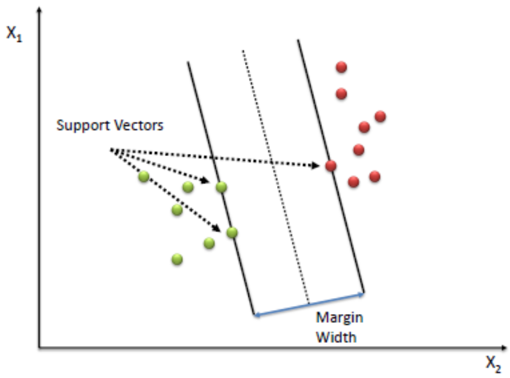
Here $sim(x, s^j)$ is the similarity between test point x and support vector s^j. Note that we have also multiplied each weighted similarity with the class of that support vector i.e. by 1 or -1, so that D is favored towards the class of that support vector.

Finally, if $D > 0$, we predict class of x to be 1, else -1.

Now having understood what SVMs are, we have following questions :

points whose weight becomes zero are not support vectors i.e. their importance is zero during test time, and rest are support vectors.

2. **So then how to learn alphas ?** Its at this point, where all the math comes. Since the question asks for layman's explanation, I will skip the math here (you can read the math here). Intuitively, we try to find that separating hyperplane, from which distance of closest training points is maximum (also known as max-margin classifier), and those closest training points then become support vectors. During the math, while optimizing the function, we get the weights of the support vectors. Pictorially, this looks like :



In the above figure, there are only 3 support vectors, so at the test time, we will compute similarity test point and only these 3 support vectors. Generally, there are very few support vectors out of all training points, and hence at test time, we need to calculate very few similarities.

3. What exactly the similarity function $sim(x, s^j)$ is ? During the math we do in above step, it turns out that the similarity function is **dot product** i.e. $sim(x, s^j) = \langle x, s^j \rangle$ (remember dot product measures similarity between two vectors).

The Fun Part : Now here comes the fun part, instead of taking dot product between two simple vectors x and s (where s is one of the support vectors), we can transform each of the vector into some other **higher dimensional** vector, and then take dot product between those two complex, higher dimensional vectors, and whole thing still works, and in fact works better since we have improved the similarity function. In particular, we can apply some transformation function ϕ to x and s to get two big vectors $\phi(x)$ and $\phi(s)$ and then we can take their dot product to get a quantity which is called **Kernel** :

$K(x, s) = \langle \phi(x), \phi(s) \rangle$

So Kernel is nothing but a similarity measure between two transformed vectors.

So our equation for computing D becomes :

$D = \sum_{j=1}^{k} y^j * \alpha^j * K(x, s^j)$

The Kernel Trick :

So now the question arises whether we need to actually transform the vectors into higher dimension to get the Kernel value ? It turns out that the answer is **not necessarily**. See, in above equation of calculating D, we require only the final answer of $K(x, s^j)$. We actually don't care, which transformation function was applied to each of the vectors. So if there is a similarity function (a kernel), which can be factorized into dot product of two transformed vectors, then that kernel function can be used in above equation. We don't even need to compute exact transformation, as long as we can prove above thing. This is called **the kernel trick**.

So earlier, we used a very simple kernel, which was just the dot product between x and s (in their original dimension). Now we may ask the question whether square of that dot product is a kernel ? i.e. whether

$K(x, s) = (\langle x, s \rangle)^2$

is a kernel or not. For simplicity, let both vectors x and s are two dimensional vectors. We can prove that it is in fact a valid kernel by noting that

$(\langle x, s \rangle)^2 = (x_1 s_1 + x_2 s_2)^2 = (x_1^2 s_1^2 + x_2^2 s_2^2 + 2 x_1 s_1 x_2 s_2)$

$= \langle \{x_1^2, x_2^2, \sqrt{2} x_1 x_2\}, \{s_1^2, s_2^2, \sqrt{2} s_1 s_2\} \rangle$

So as we see here, this kernel is a dot product between two vectors :

which are nothing but transformation of vectors x and s.

Note that we don't have to actually calculate transformed vectors, we just require kernel function, which is very easy to compute, for example, here, its just the square of dot product.

4. Which kernel function to use ? We have 3 kinds of common kernel functions :

- **Linear Kernel :** $K(x, s) = \langle x, s \rangle$. This is just the dot product between original vectors.
- **Polynomial Kernel :** $K(x, s) = (\langle x, s \rangle)^p$. We saw an example of this above, in which $p = 2$.
- **Gaussian Kernel :** $K(x, s) = \exp(-||x - s||/\sigma)$. This is just an unnormalized gaussian function (we don't need to normalize it, since we need just similarity measure, and not a distribution). This is super powerful kernel since it is the dot product between two infinite dimensional vectors. So the amazing part is that we are calculating dot product between two infinite dimensional vectors without actually having those vectors !!!.

So to summarize, SVM is the generalization of NN algorithm with weights assigned to some of the training instances (support vectors) which decide the class of test instance. Moreover, instead of using simple similarity functions, we can apply a variety of complex similarity functions.

There is a lot of math behind it, but I think you get the big picture.

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Other answers so far explain what support vectors are.

To find the support vectors, you need to solve an optimization problem.

To be precise, it is a linear optimization problem. The simplest method of solving it is the simplex algorithm. This can be inefficient for sizable problems, so there are many more efficient approximations, as well as efficient algorithms for problems where certain assumptions hold that allow for more efficient computation.

the conception of the zygote, the first cell of existence. The Soul is a Spark Of Unique Life. The Soul creates life in the very first cell. When we are fertilized in the womb of our mother, it is with that cell, the first cell which has the Soul, the Spark Of Unique Life, that the cells multiply from one to two to four, and then the foetus is developed. It is wrong to believe that the foetus is first created... (more)

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It is simple if you are using a quadratic solver(in most cases we use it). The vector returned from quadratic solver be $\alpha = (\alpha_1, \alpha_2, \ldots\ldots, \alpha_N)$ where N is the number of training examples used to train SVM.

This vector will have non-negative values ($\alpha_i \geq 0 \quad \forall i = 1\, to\, N$) and out of these many will be zero. We are only interested in positive values because x_i corresponding to those non zero α_i forms the support vectors of our hyperplane where $(x_i,\ y_i)$ is a training example.

EDIT: The best learning resource for SVM online is

1.

2. ... (more)

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Related **How do we select which data points act as support vectors for SVM?**

For a given data set, you cannot select them and still call the classifier an SVM. First I'll say why, and then how to try to dictate support vectors if possible. If you understand SVMs, you can skip the next para.

The support vectors just come out of the wash from the desire to maximize the margin. An SVM classifier can be thought of as an infinite `plank' of unknown thickness that needs to be slipped between the two data sets. Maximizing the margin is maximizing the thickness of that plank, and so the plank will touch some of the data points when it is thickest. If it didn't touch any points,... (more)

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I'm just gonna give an analogy since the already provided answers are pretty spot on.

Think of it as a really buff guy hitting the gym to maximize his muscle gains. His sole goal in the near future is to walk sideways through doors and shit.

When you get buff enough, you'll adjust your walking style though doorways and narrow passages by maximizing available space on both your left and right. SVM work in the same way.

SVM is like the most buff guy at the gym, who always takes a nod-worthy sideway walk through every doorway. *Not because he intends to*, but because it's *the most sure way to avoid hi*... (more)

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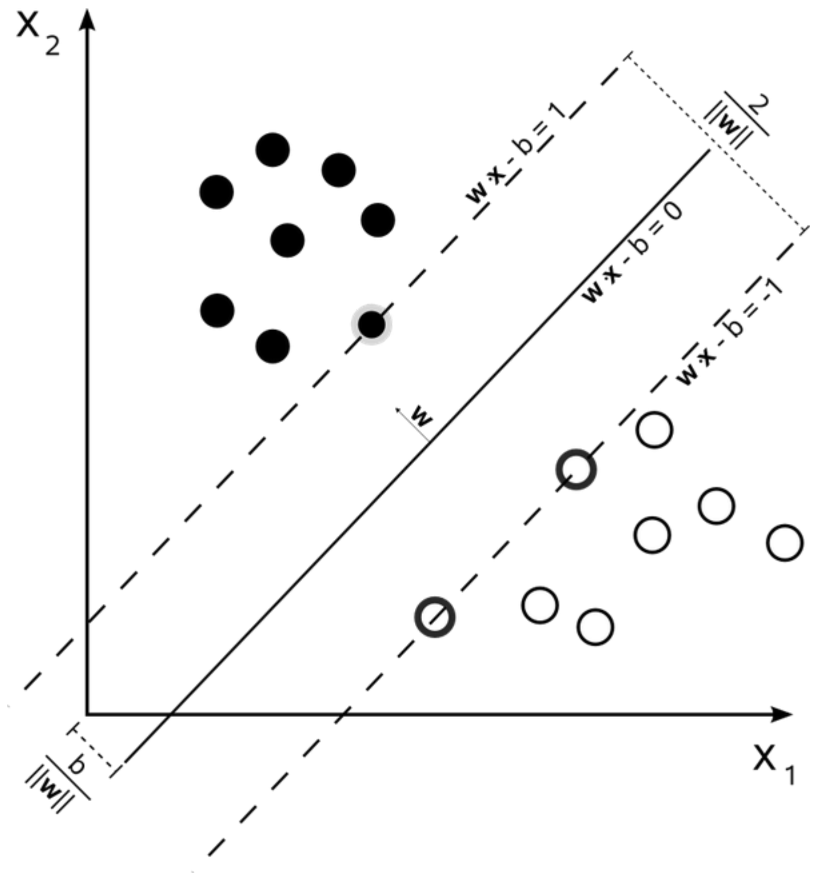
Support vector machines are *maximum-margin classifiers*, which means they find the hyperplane that has the largest perpendicular distance between the hyperplane and the closest samples on either side. The closest samples on either side are the support vectors.



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svm which means support vector machine wants the plane margin as maximum and uses two planes as margin one is pi+and anotger one is pi- . And it uses the central plane to classify the points.coming to question how it determines its support vectors it uses smo algorithm such that there is value name alpha these are determined using smo.alphas becomes zero for all points except for support vectors.ACTUALLY WHAT HAPPENS IN SUPPORT VECTOR MACHINE IS WE WILL FIND THE PLANE PI PLUS AND PI MINUS USING SMO ALGORITHM LATER PI IS DETERMINED FROM THOSE TWO.AND LA5ER CLASSIFICATION CAN BE DONE.

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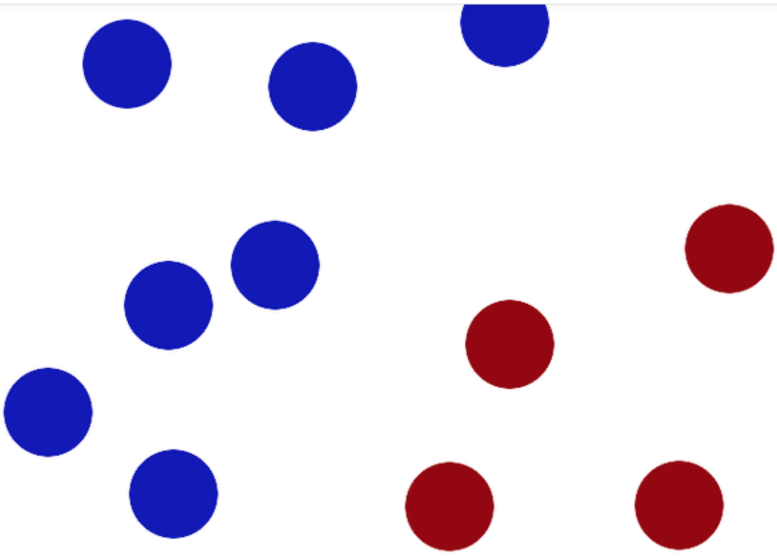
Related **What does support vector machine (SVM) mean in layman's terms?**

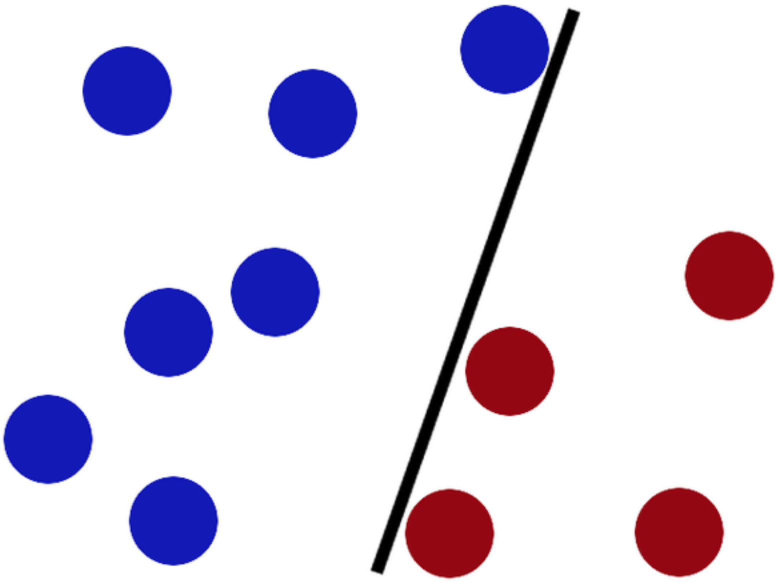
Found this on Reddit: Please explain Support Vector Machines (SVM) like I am a 5 year old. • /r/MachineLearning

Simply the best explanation of SVM i ever found.

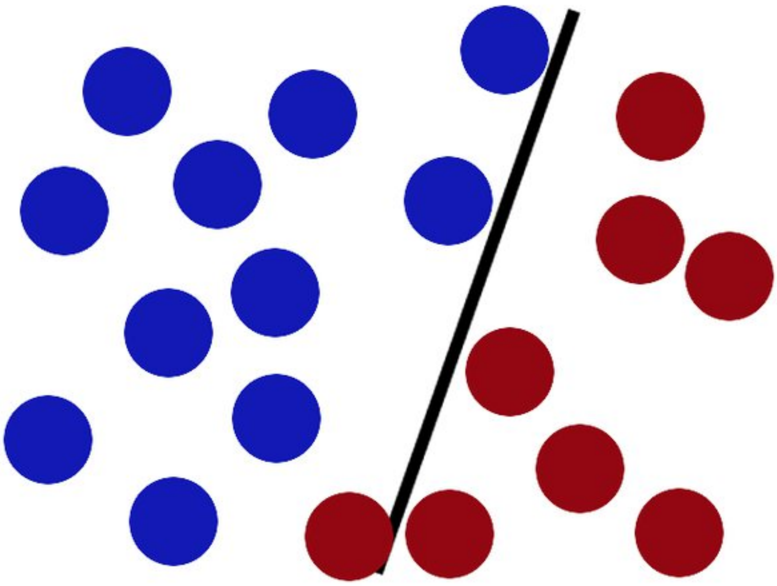
We have 2 colors of balls on the table that we want to separate.



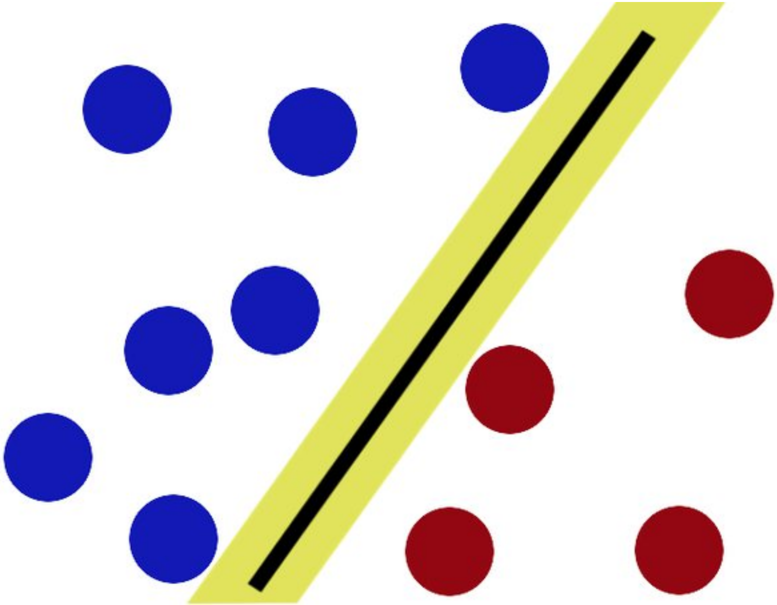
We get a stick and put it on the table, this works pretty well right?



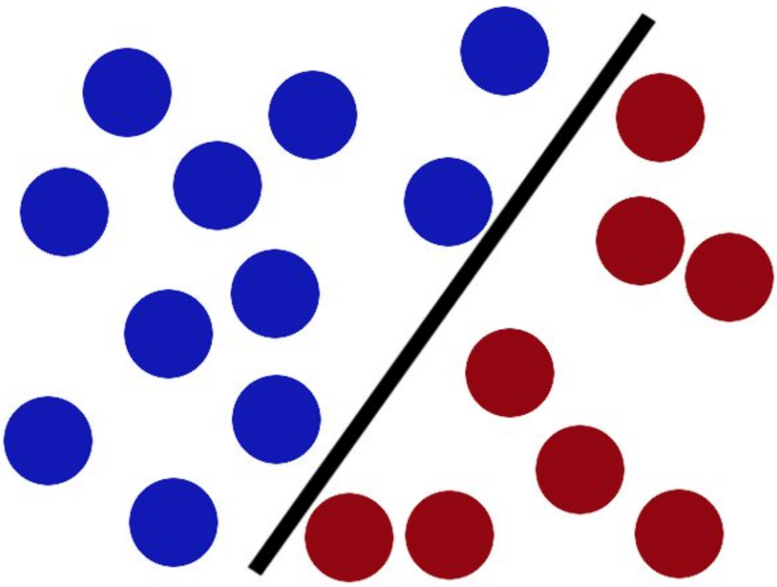
Some villain comes and places more balls on the table, it kind of works but one of the balls is on the wrong side and there is probably a better place to put the stick now.



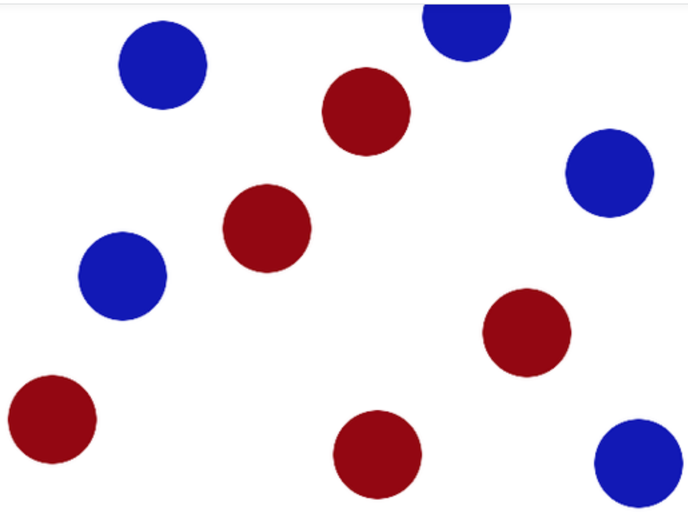
side of the stick as possible.



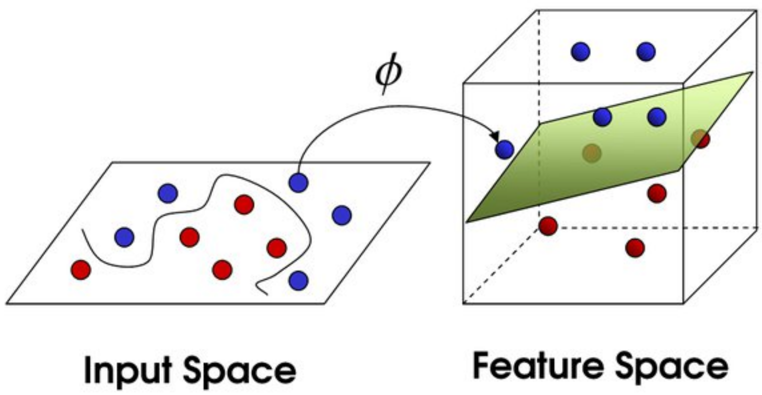
Now when the villain returns the stick is still in a pretty good spot.



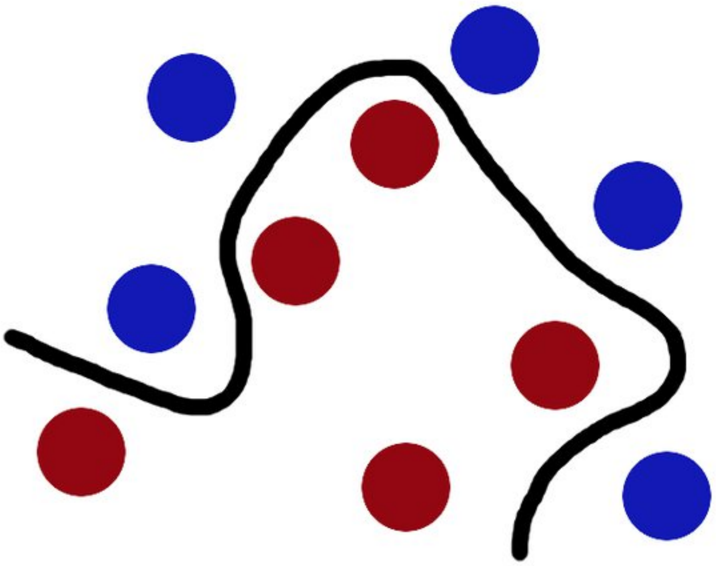
There is another trick in the SVM toolbox that is even more important. Say the villain has seen how good you are with a stick so he gives you a new challenge.



There's no stick in the world that will let you split those balls well, so what do you do? You flip the table of course! Throwing the balls into the air. Then, with your pro ninja skills, you grab a sheet of paper and slip it between the balls.



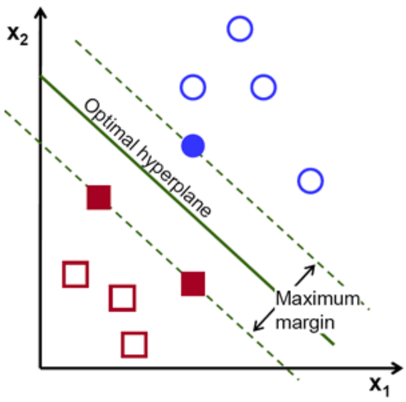
Now, looking at the balls from where the villain is standing, they balls will look split by some curvy line.



Boring adults the call balls data, the stick a classifier, the biggest gap trick optimization, call flipping the table kernelling and the piece of paper a hyperplane.

One other point that I like to mention about the SVM is how it is defined by the boundary case examples. (CS109 course:SVM Explanation). Picture source: OpenCV documentation, Willow Garage

Assume you want to separate Apples from oranges using SVM. The Red square are Apples and the blue circles are oranges.



Now see the support vectors(The filled Points) which define the margin here.

Intuitively the filled Blue circle is an Orange that looks very much like an Apple. While the filled Red squares are Apples that very much look like oranges.

Think for this another time. If you would want your kid to learn to differentiate between an apple and orange you would show him a perfect apple and a perfect orange. But not SVMs they want to only see an apple that looks like an orange and vice versa. This approach is very different from how most Machine learning algorithms operate, and maybe thats why it works so well in some cases.

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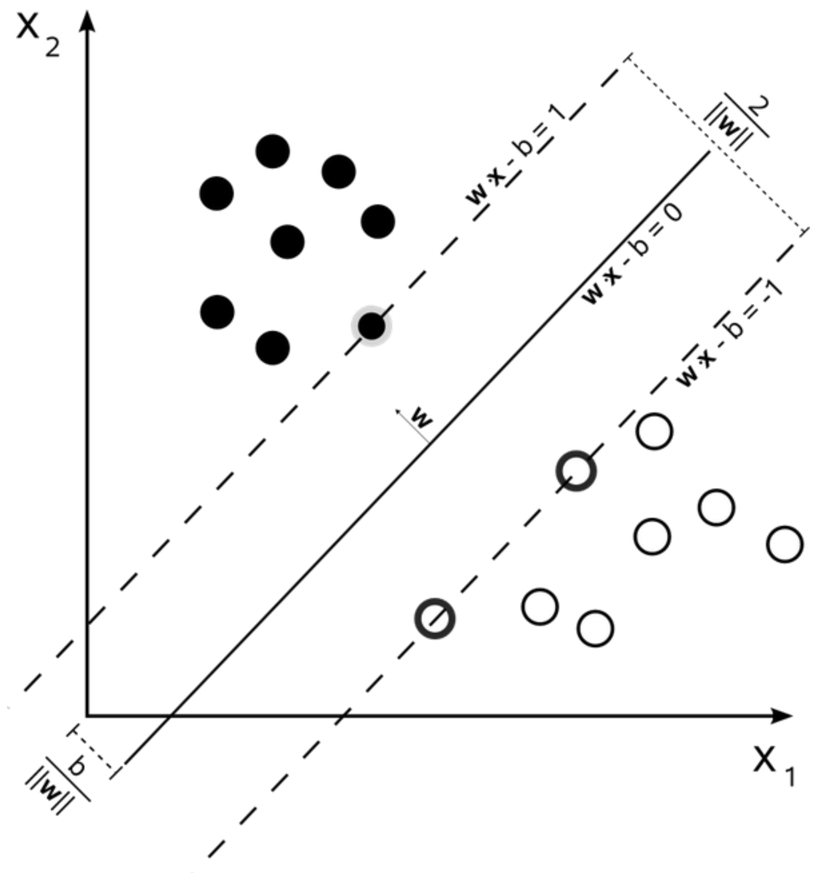
Originally Answered: what is the purpose of support vector in SVM?

A support vector machine attempts to find the line that "best" separates two classes of points. By "best", we mean the line that results in the **largest margin** between the two classes. The points that lie on this margin are the *support vectors*.



Here we have three support vectors.

The nice thing about acknowledging these support vectors is that we can then formulate the problem of finding the "maximimum-margin hyperplane" (the line that best separates the two classes) as an optimization problem that only considers these support vectors. So we can effectively throw out the vast majority of our d... (more)

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A2A

1] **Sparse Representation in Non Linear Space**

In SVM, our attempt is to find a linear function $f(x) = w^\top x + b$, such that $\text{sgn}(f(x))$ will give us the label of x. So, what we are looking to do is to estimate w.

This is essentially what we do in the primal form of SVM:

$\min \|w\|^2 + C\sum\xi_i$

s.t. $y_i(w^\top x_i + b) \geq 1 - \xi_i$ for all i where α_i are the dual variables and they are non-zero only for those points that lie on or inside the margin.

But, $f(x) = w^\top x + b$, substituting for w we get,
$f(x) = \sum_{i=1}^{n} \alpha_i y_i x_i^\top x + b$

This tells us that we can represent our linear classifier in terms of the points that lie on the margin or within. Excellent, but what are we gaining because of this?

Note that the function f(x) is now represented only using dot product between the terms. So, not only is the dual of SVM uses dot product but the classifier representation itself is in terms of dot product. This enables us to use non-linear classifiers and compute classification hyperplanes in large dimensional spaces. Since only the points on (and inside) the margin have non-zero dual variable value, it also implies that the classifier has a sparse representation.

2] **Gives a computable upper bound on SVM error**
It is amazing that these points lying on the margin not only dictate the hyperplane but also provide a way to compute tighter bounds on the error made by SVM via the leave one out procedure.

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If you understand the math formula of SVM, decision of SVM = $\boldsymbol{W}'\boldsymbol{x}+b$

In here, we don't consider about *b*, so the parameter of SVM is ***w***, the solution of ***w*** is shown as below

$$\frac{\partial L(\boldsymbol{w}, b, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\xi})}{\partial \boldsymbol{w}} = 0$$

$$\Rightarrow \frac{\partial\left[\frac{1}{2}\boldsymbol{w}^T\boldsymbol{w} + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left(y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{n}\beta_i\xi_i\right]}{\partial \boldsymbol{w}} = 0$$

$$\Rightarrow \boldsymbol{w} - \sum_{i=1}^{n}\alpha_i y_i \boldsymbol{x}_i = 0 \Rightarrow \boldsymbol{w} = \sum_{i=1}^{n}\alpha_i y_i \boldsymbol{x}_i$$

xi is the training data.

you use soft-margin SVM, then some of data would be allowed out of margin, depend on your penalty term (usually represent by C).

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As many as needed.

There is no single answer because the number of support vectors will depend on the kernel choice, the dimensionality of the problem, the noisiness of the data and the parameter settings of the algorithm.

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So we can formulate the primal optimization problem of the SVM as:

$$\underset{w}{min}\ \|w^2\| + c\sum_{i=1}^{n}\xi_i$$

s.t.

$$y_i(w^Tx_i + b) \geq 1 - \xi_i \text{ for } i = 1, \dots, n$$

Let's just parse that for a moment. First, let's ignore all terms to do with ξ. w is our linear classifier, i.e.,:

$$f(x) = w^Tx + b$$

is our classification function (1 if greater than 0, 0 otherwise). Maximizing its norm is the same as maximizing the margin of classification (hence SVM is a max-margin classifier). The constraints (without the ξ terms) mean that we have to get every single datapoint correctly classified. Basically, without ξ's this is saying find the max-margin classifier that perfectly separates the training data.

Obviously, not all data is linearly separable (even if we use kernels to map to higher dimensional spaces). To not break the SVM every time the data is not linearly separable, we add the ξ terms, which soften the constraint, and allow for some training examples to be incorrectly classified. c controls the tradeoff of how well separated our data is and the size of the margin. That is enough in general, but we can definitely take this further.

$$w = \sum_i \alpha_i y_i x_i$$

Substituting this in for $f(x)$:

$$f(x) = \sum_i \alpha_i y_i (x_i^T x) + b$$

We are beginning to touch upon the beauty of SVMs. This formulation implies that we can classify a new example x based on the dot product with our training data. How to find these pesky α_i 's though, and what do they mean? Let's rewrite $||w^2||$ in our new form:

$$||w^2|| = \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j) + b$$

This leads to the dual form of the SVMs:

$$\underset{\alpha \geq 0}{w} \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

s.t.

$$0 \leq \alpha_i \leq c$$

$$\sum_i \alpha_i y_i = 0$$

This (convex!) optimization problem can be solved using quadratic programming, which is simple enough. The α terms can be interpreted as support vectors - i.e., $f(x)$ only depends on training examples for which $\alpha_i > 0$, which in practice is usually a sparse subset of the data. Finally, since we can express the optimization problem and classification function in terms of dot products with the training data, SVMs lend themselves naturally to kernels, which allow us to perform linear classification in high (potentially infinite) dimensional spaces!

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In a support vector machine (SVM), support vectors are the data points that lie closest to the decision boundary (or hyperplane) separating the classes.

The distance between these support vectors and the hyperplane is called the margin. The goal of SVM is to maximize this margin, so that the hyperplane with maximum margin is called the optimal hyperplane.

To compute the maximum margin, SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors.

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One-vs-all and one-vs-one are two approaches to multiclass classification using support vector machines (SVMs).

In the one-vs-all approach, a separate binary SVM classifier is trained for each class, with the goal of predicting whether a sample belongs to that class or not. When you want to predict the class of a new sample, you feed it to all of the classifiers and choose the class that is predicted most often.

In the one-vs-one approach, a separate binary SVM classifier is trained for each pair of classes. For example, if you have three classes, you would train three classifiers: one for class… (more)

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The equation of the separating hyperplane for the SVM under kernel K is

$$\sum \alpha_i y^i K(x^i, x) + b$$

where by KKT conditions imply $\alpha_i != 0$ for only the SV's. From the equation it should be evident if the SV's change the hyperplane would almost always shift .

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I assume you have an analytical expression for the separating hyperplane, represented as an equation $f^*(x) = 0$.

To check if the empirical hyperplane follows the theoretical one, observe that an SVM learns a function $f(x)$ that assigns label +1 or -1 to the entire input space, with the separating hyperplane represented by $f(x) = 0$. So in principle, you can check on the entire surface of the true hyperplane the value of $f(x)$, and it should ideally be 0 or in practice, very close to zero. Checking on the entire surface is clearly infeasible in practice, so you can sample points on the surface to approximately find the quality of the solution.

One caveat: You should only expect $f(x)$ to be close to $f^*(x)$ where the density of training points is high enough. For regions where there is little or no data, the two may differ significantly. And this is true for any machine learning model, not just SVMs. So when you sample points on the surface and take an average difference between $f(x)$ and $f^*(x)$, sparse regions will dominate the result in a naive implementation, and you will almost always conclude that $f(x)$ and $f^*(x)$ do not match. So you must take some sort of weighted average of the samples where the weight captures the density of training instances around the sample point, or choose sample points in such a manner that you only sample in high density regions.

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Related **Would the support vectors in the SVM algorithm change with the scaling of the functional margin?**

Originally Answered: Would the support Vectors in the SVM algorithm change with the scaling of the functional margin ?

A2A

>> The entire algorithm is then derived by setting i.e. Gamma(hat) to 1

That is not exactly true, although that's certainly one way to think about it. The algorithm proceeds by replacing w and b with new variables $w = \hat{\gamma}w'$ and $b = \hat{\gamma}b'$. You will notice that when you do these substitutions, $\hat{\gamma}$ will disappear.

So, in this new formulation you are solving for w' and b' instead of w and b. Now, given this information, you can make two observations:

1. You can obtain the solution for the original formulation for any $\hat{\gamma}$ from the new formulation by scaling w' a

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SVM finds a gap between the data of two classes, such that the gap is as wide as possible and then new examples are classified based on which side of the gap it belongs. This can be imagined as a linear boundary in two-dimension which decides to which class the example belongs.

For non-linear classification, the input is mapped to higher dimention using some function (kernel), so that the classes can be separated by a linear plane in higher dimension space which is again mapped to original state space in lower dimension in the form of non-linear boundary. This non-linear boundary is used for cl... (more)

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A support vector machine (SVM) is a type of supervised learning algorithm that can be used for classification and regression tasks. It works by finding the hyperplane in a high-dimensional space that maximally separates different classes.

An SVM is trained by providing a set of labeled training examples, which are pairs of input data and corresponding target labels. For example, in a classification task, the input data might be a set of features, and the target labels might be the class labels (e.g., "red," "green," "blue"). The SVM uses these training examples to learn the optimal hyperplane t... (more)

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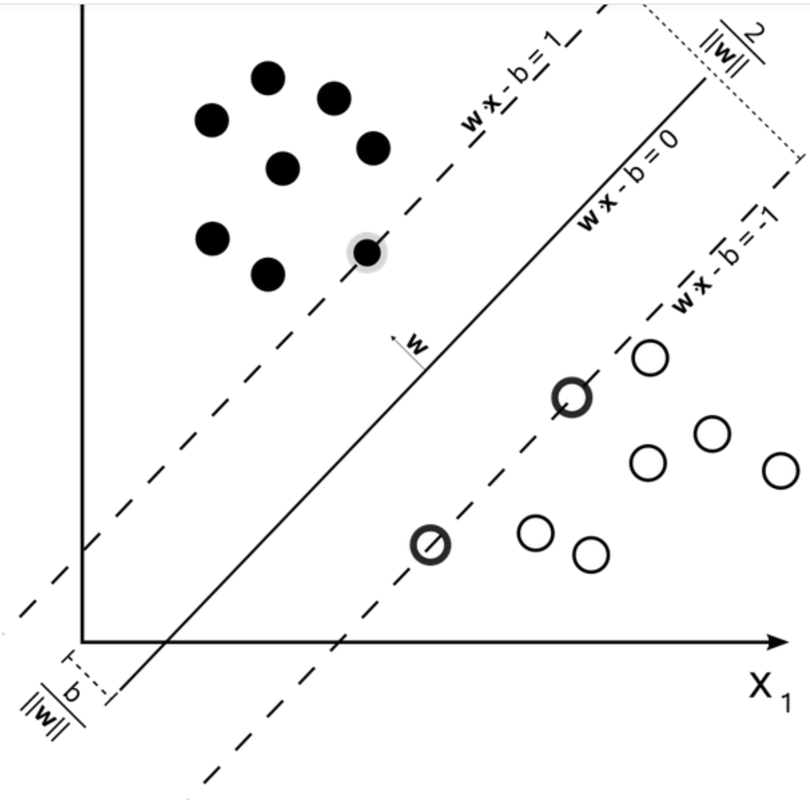

A vanilla SVM is a type of linear separator. Suppose we want to split black circles from the white ones above by drawing a line. Notice that there are an infinite number of lines that will accomplish this task. SVMs, in particular, find the "maximum-margin" line - this is the line "in the middle". Intuitively, this works well because it allows for noise and is most tolerant to mistakes on either side.

That's all a (binary) SVM really is. We draw a straight line through our data down the middle to separate it into two classes. But what if we don't want a straight line but a curved one? We achieve this not by drawing curves, but by "lifting" the features we observe into higher dimensions. For example, if we can't draw a line in the space (x1, x2) then we may try adding a third dimension, (x1, x2, x1 * x2). If we project the "line" (actually called a "hyperplane") in this higher dimension down to our original dimension, it looks like a curve. This video is an excellent illustration:

By using the kernel trick, we can do these lifts very efficiently (actually, we don't really do them at all). Kernels allow us to draw "straight lines" in very high dimensions, and even infinite dimensions.

EDIT: notice that to draw the maximum margin line, only the data points closest to the line matter (these are the ones with darker borders). These points are known as "support vectors" which gives SVMs its name.

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Related **Knowing the SVM's support vectors, can I find the weights (importance) to my features?**

If you already know the support vectors, then you can find out the decision boundary easily. Of course you can find out the weights!

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Related **What are the disadvantages of support vector machines (SVMs)?**

A few of the disadvantages of SVMs are as follows:

1. **Computational Complexity:** *SVMs can be computationally expensive during training, especially when dealing with large datasets.*
2. **Sensitive to Outliers:** *SVMs are sensitive to outliers and can be affected by the presence of noisy or irrelevant data.*
3. **Limited to Smaller-Scale Problems:** *SVMs can be less effective for larger datasets and problems with a high number of features.*
4. **Difficulty in Model Selection:** *It can be difficult to choose the right kernel function and the optimal parameters for the SVM model.*
5. **Requires Data Preprocessing:** *SVMs require data pr*

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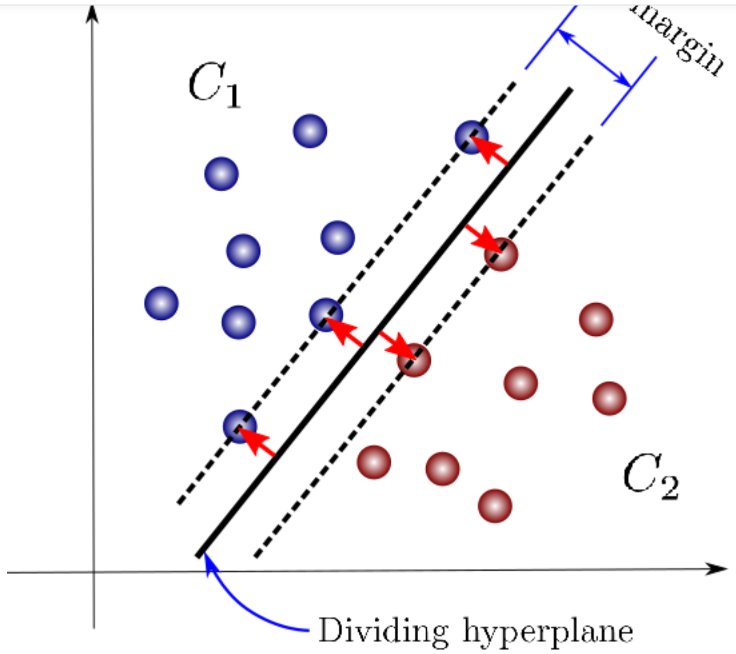
SVM can be used both for regression and classification. It is particularly useful when the data is non-linear.

We can use SVM when the number of attributes is high compared to the number of data points in the dataset.

SVM uses a technique called Kernel Trick which projects the data onto the higher dimensions so as to make it easier to classify the data.

It creates the decision boundary that can classify n-dimensional space into classes for classification. The best boundary is called a hyperplane.



Hyperplane dividing the n-dimensional space (Source: mlfactor.com)

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Related **What is the difference between kernelized and linear support vector machines (SVM)?**

Support Vector Machines (SVMs) are a type of supervised machine learning algorithm used for classification and regression problems.

Linear Support Vector Machines (LSVMs) are a specific type of SVM that operate on a linear boundary to separate the data into different classes. The boundary is a line or a hyperplane in high-dimensional space that maximizes the margin between the classes. LSVMs are best suited for linearly separable data.

Kernelized Support Vector Machines (KSVMs), on the other hand, are a more general type of SVM that allow for non-linear boundaries to separate the data. KSVMs use… (more)

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Originally Answered: When training a SVM classifier, is it better to large or small number of support vectors?

That's a very general question that can't be answered specifically. If you e.g., a RBF kernel SVM you'd typically end up with more support vectors than in a linear model. If you data is linearly separable, the latter may be better, if not, the former may be better.

Also, you have to differentiate between computational efficiency and generalization performance. If you have more vectors, your classification becomes more expensive for example, especially in kernel SVM where you have to recalculate the distances between every new sample and the entire training set.

I recommend to validate your mode... (more)

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In general the smaller the number of support vectors the more likely will you be getting a more generalizable classifier.

To understand this see:
Page on microsoft.com

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E(P(error))>= E(# SVs)/#Samples

Essentially, the term on the left is the genralization error and the right talks about the compression achieved. In general, machine learning algorithms that achieve higher compression are more likely to be generalizable, and in a loose sense, better.

In the SVM #SVs measures compression in a certain sense. Now, the statement is probablistic, which means that i... (more)

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Related **What is your favourite kernel to use with a support vector machine (SVM)?**

When I use support vector machines (SVM) it is usually a linear SVM feeding on high-level features at the end of a model.

I don't really like using the kernel trick, I think it is better to have a much more powerful feature extractor like a pre-trained convolutional neural network (CNN) drive a linear SVM than use the shallow kernel trick.

The kernels may find a mapping in high dimensional space, which is cool and all, but it might overfit and not generalize as well the above mentioned approach.

Hope this helps.

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Related **When training an SVM classifier, is it better to have a large or small number of support vectors?**

Many of the most popular svm implementations fail by accruing computational errors for having many support vectors. Because of this, it's not practical to have a too large dataset.

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An SVM is a statistical learning method most commonly used to perform classification and regression tasks.

that the optimal hyperplane is the one that has the maximum margin (distance) between the classes. This results in a decision boundary.
The SVM is also part of a larger group of learning methods known as kernel methods. This means that we can make use of a special type of function known as a kernel function to ... (more)

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